

Data Flow Diagram

Date	15 March 2026
Team ID	SWTID-2026-2085
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to illustrate how data moves through the OptiCrop system. The diagram should show how farmers provide soil and environmental information, how the system processes this data using a machine learning model, how it interacts with the crop dataset, and how the recommended crop is returned to the farmer. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Represents an external user or source that sends or receives data, such as the Farmer or Weather Data Source .
Rectangle with numbered header	Process	Represents a processing activity, such as Input Soil Data, Validate Data, Crop Prediction, or Generate Recommendation .
Rectangle (solid fill, no number)	Data Store	Stores information used by the system, such as the Crop Dataset, Soil Data, or Prediction Results Database .
Labeled arrow	Data Flow	Represents the movement of data between the farmer, system processes, and data stores (e.g., Soil Data, Weather Data, Crop Recommendation).

Example

The diagram below illustrates how data flows through the **OptiCrop: Smart Agricultural Production Optimization Engine**, showing the interaction between the farmer, the crop recommendation system, the machine learning model, and the crop database to generate accurate crop recommendations.

